

CROP ECOLOGY, PRODUCTION & MANAGEMENT

Assimilatory Capacity Effects on Soybean Yield Components and Pod Number

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ABSTRACT

Soybean [*Glycine max* (L.) Merr.] yield is more restricted by assimilatory capacity (source strength) during the reproductive (R1 to R7) compared with vegetative period (emergence to R1). Although pod number (per unit ground area) is recognized as an important factor affecting yield, the period in which this yield component is source restricted has not been clearly identified. Therefore, the objective of this study was to determine this period by identifying yield components through which source strength affects pod number and finding when these yield components are determined. Field studies were conducted during 1991 and 1992 with 'Centennial' soybean at Baton Rouge, LA, on a Mhoon silty clay soil (fine-silty, mixed, nonacid, thermic Typic Fluvaquent). Treatments were partial defoliations designed to create light interception differences during R1 to R7. Yield was reduced 23% by defoliation. Harvest index, which was 0.59 for the control, ranged from 0.56 to 0.60 for the defoliation treatments. Source strength influenced pod number (per unit ground area) through branch dry matter, branch number, branch node number, and pods per reproductive node on the whole plant. Pods per reproductive node were regulated by pod initiation (pods at least 0.5 cm long) and/or abortion of initiated pods. Determination of final pod number occurred at 10 to 12 d after R5. In conclusion, pod number was source restricted from R1 to 10 to 12 d after R5. Stresses that restrict source strength during this period should be avoided to optimize pod number and yield.

GENETIC AND CULTURAL STRATEGIES for increasing soybean yield can be improved by identifying growth periods where potential yield is limited by assimilatory capacity (source restricted). Such knowledge can be gained by determining yield component responses to source strength and knowing when these components are formed. This information can be used to select traits for incorporation into current cultivars and to make recommendations to farmers of when optimum assimilatory capacity is necessary for highest yield. Previous research (Brun, 1978; Christy and Porter, 1982; Schou et al., 1978) has indicated that yield is more influenced by changes in source strength during the R1 to R7 period compared with the emergence to R1 period (stages according to Fehr and Caviness, 1977). Jiang and Egli (1993) cited several studies (Egli and Yu, 1991; Johnston et al., 1969; Schou et al., 1978; Shibles et al., 1975) indicating that seed number (per unit ground area) was responsive to altered environmental conditions during flowering and pod set. Because seed per pod is genetically influenced (Shibles et al., 1975), seed number (per unit

ground area) is usually highly correlated with pod number (per unit ground area). However, yield component mechanisms through which source strength influences pod number have not been completely identified. In response to a light interception (LI) gradient during R1 to R7, pod number was regulated by pods per reproductive node (Board and Harville, 1993), thus demonstrating source restriction during pod formation (R3 to R5). In their study, reproductive node number was constant across treatments.

Pod number is determined by pods per reproductive node and reproductive node number. The first parameter is determined by the difference between total pods per reproductive node initiated [pods at least 0.5 cm long (Fehr and Caviness, 1977)] and those aborted. Reproductive node number is determined by total node number and the percentage of these nodes becoming reproductive (percentage reproductive nodes). Although main stem node number in determinate cultivars is determined during the vegetative period, all other yield components contributing to pod number are formed mainly during R1 to R5 and possibly into the late reproductive period (R5 to R7). Contribution of each of these yield components to final seed yield can be summarized by the following equations:

$$\text{seed yield (g m}^{-2}\text{)} = f[\text{pod number (no. m}^{-2}\text{)}]$$

$$\begin{aligned} \text{pod number no. m}^{-2} &= \\ &\text{pods per reproductive node (no.)} \times \\ &\text{reproductive node number (no. m}^{-2}\text{)} \end{aligned}$$

$$\begin{aligned} \text{pods per reproductive node (no.)} &= \\ &\text{total pods initiated per reproductive node (no.)} - \\ &\text{total pods aborted per reproductive node (no.)} \end{aligned}$$

$$\begin{aligned} \text{reproductive node number (no. m}^{-2}\text{)} &= \\ &\text{total node number (no. m}^{-2}\text{)} \times \\ &\text{nodes becoming reproductive (\%)} \end{aligned}$$

$$\begin{aligned} \text{node number (no. m}^{-2}\text{)} &= \\ &\text{main stem node number (no. m}^{-2}\text{)} + \\ &\text{branch node number (no. m}^{-2}\text{)} \end{aligned}$$

$$\begin{aligned} \text{branch node number (no. m}^{-2}\text{)} &= \\ &f[\text{branch number (no. m}^{-2}\text{)}] \text{ and} \\ &f[\text{branch dry matter (g m}^{-2}\text{)}] \text{ (Board et al., 1990)} \end{aligned}$$

The objective of this study was to determine responses of these yield components to altered source strength to identify growth periods when potential pod number and yield were source restricted.

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MATERIALS AND METHODS

Culture

Field studies were conducted at the Ben Hur Farm near Baton Rouge, LA (30° N Lat). Cultivar Centennial (Maturity Group VI) was planted at a high seeding rate on 13 July 1991 and 15 July 1992. Late planting was used because this test was part of an ongoing research effort to study soybean growth responses at late plantings in the southeastern USA. Due to weather and double cropping with winter cereals, late planting is frequent in the region. Shortly after emergence, plants were thinned to a density of 325 000 plants ha⁻¹. Stand counts were made at maturity to verify plant densities within each treatment. Based on soil test recommendations, fertilizer was incorporated into the soil before planting at 0-41-78 kg ha⁻¹ (N-P-K). Row width and length were 50 cm and 6.1 m, respectively. Row orientation was east to west. Experimental units were eight consecutive rows. Plants were supported at R1 by nylon mesh netting (12.7 by 12.7 cm) stapled to wood stakes to prevent lodging. Weeds, diseases, and insects were suppressed by recommended pesticides.

Experimental Design

Experimental design was a randomized complete block with four replications and 2 yr as blocking factors. The treatments were a control (no defoliation) and four defoliation treatments (DF) during the reproductive period. Partial defoliation (Board and Harville, 1993) was used to induce LI changes, because defoliation effects on yield have been shown to be directly caused by reduced LI and canopy photosynthesis (Ingram et al., 1981). A defoliation treatment consisted of maintaining the area 12.5 cm to both sides of the interrow midline (i.e., 50% of the interrow area) free of leaf blade material until R3 (DF-R3), R4 (DF-R4), R5 (DF-R5), or R6.5 (DF-R6.5). Marking strings attached to small stakes were stretched alongside the rows at distances of 12.5 cm from the interrow midline. Parts of leaves that extended over the string into the 50% interrow area during the defoliation period were removed with sharp scissors twice each week. Once the respective developmental period was reached for a given treatment (either R3, R4, R5, or R6.5), defoliation was discontinued and leaf growth allowed into the middle 50% interrow area.

Data Collection and Analyses

Light interception was measured at 38, 47, 53, 60, 66, 73, 80, 87, and 93 d after emergence (DAE) for all treatments. Determinations were made with a 1-m long LI-COR Line Quantum sensor (LI-COR, Lincoln, NE) connected to a LI-1000 data logger. Photosynthetically active radiation was measured in micromoles per square meter per second at ground level, starting adjacent to the plant row and then at 12.5-cm intervals across the interrow space to the next plant row. Flux measurement at canopy top was then made and the LI percentage determined. All recordings were made parallel to the rows and taken between 1200 and 1300 h in full-sun conditions. Data were analyzed by analysis of variance with DAE a split plot within defoliation treatments (main plots). Mean separation was done with LSD using appropriate error terms.

Pods per reproductive node were determined for all experimental units at 49, 50, 51, 53, 59, 67, 81, and 93 DAE according to the following method: (i) Five plants per plot were randomly selected at each sampling date. (ii) For sampling dates 49 through 59 DAE, total viable pod number included

all green pods >0.5 cm. No yellow aborted pods were evident during this period. On subsequent sampling dates, pod counts included only pods that manifested enlargement of the pod cavity. Several small pods (<4 cm) remained on the plants during 67 to 93 DAE but showed no enlargement and were counted in a separate category from viable pods. Comparison of pod counts throughout the 67 to 93 DAE period indicated these small pods remained inviable and did not contribute to the economic yield of the plant. (iii) Total number of reproductive nodes was then determined. A reproductive node was any node that had at least one viable pod. (iv) Pods per reproductive node were then determined by dividing total viable pod number by number of reproductive nodes. Statistical analysis of pods per reproductive node was by analysis of variance with sampling dates being split plots within each defoliation treatment (main plots). Mean separation was according to least significant difference (LSD).

Pod abortion percentage was calculated by subtracting final pods per reproductive node from maximal pods per reproductive node, dividing the difference by maximal pods per reproductive node, and multiplying by 100. Maximal pods per reproductive node number were the average of counts at 51 and 53 DAE. By these dates, all treatments had initiated their maximal pod number and no abortion had occurred. Final pods per reproductive node were the average of counts at 67, 74, 81, and 93 DAE, because no significant change occurred for any treatment during 67 to 93 DAE. This method does not include flower abortion but is aimed at determining the percentage of initiated pods which abort.

Ten plants per plot were randomly selected at maturity for yield component analysis. Plants were separated into main stems and branches, and analyzed for total branch length, branch number, node number, reproductive node number (node bearing at least one pod having at least one seed), percentage reproductive nodes (percentage of total nodes becoming reproductive), pods per reproductive node, pod number, seed per pod, seed number, and seed size. Partition of reduced sample seed yield of defoliation treatments vs. control into seed number and seed size was mathematically accomplished. Seed yield, adjusted to 130 g kg⁻¹ moisture, was measured on two interior rows for each plot that had been end trimmed to 4.3 m for harvest. Dates of emergence, R1, R3, R4, R5, R6.5, and R7 were based on biweekly random sampling of five plants per plot. Pod abortion percentage, all yield components, and seed yield were subjected to analysis of variance and mean separation by LSD.

RESULTS AND DISCUSSION

Light Interception Effects on Yield and Yield Components

Analysis of variance demonstrated that defoliation treatment significantly affected LI ($P = 0.0001$) at each sampling date (data not shown). Significant year $\times$ treatment interactions occurred only at 60 DAE ($P = 0.01$) and 87 DAE ($P = 0.05$). Therefore, LI was averaged across years for each sampling date (Fig. 1). Significant differences between control vs. all defoliation treatments occurred by 38 DAE. By R5, the control reached 96% LI, whereas defoliation treatments had only 75 to 85% LI (Fig. 1).

Within each year, yield declined as defoliation treatment was lengthened (Tables 1 and 2). Harvest index also declined as defoliation treatment was lengthened

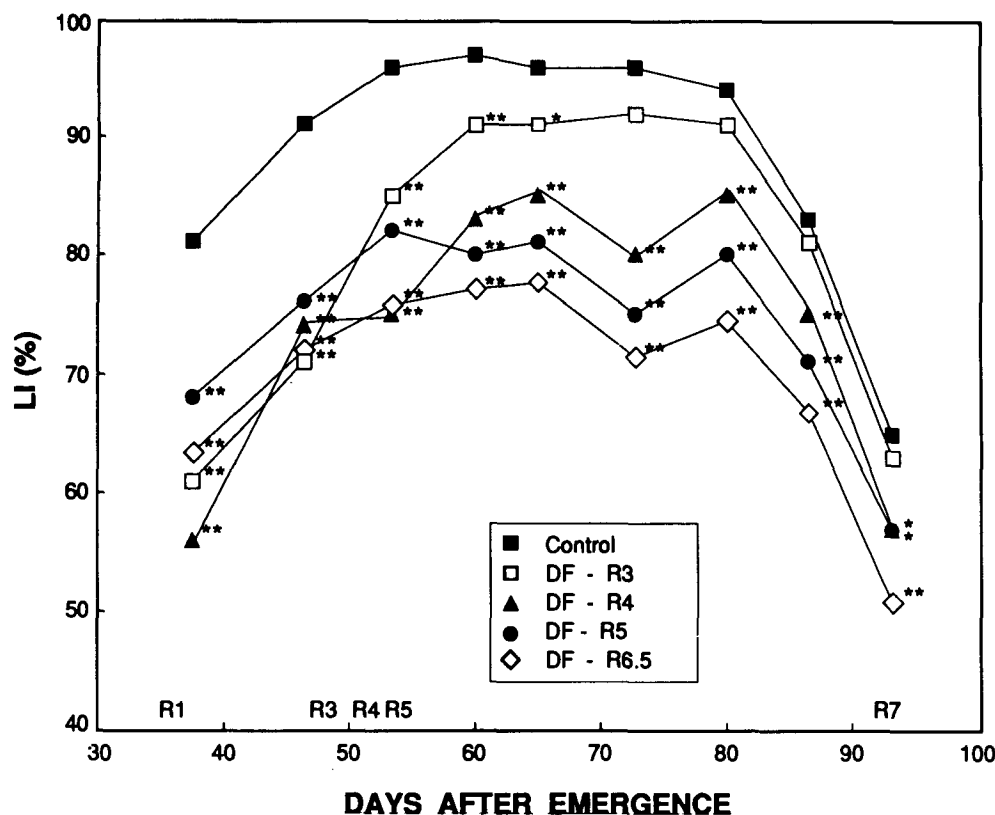


Fig. 1. Light interception (LI) of soybean as affected by defoliation treatments (DF), averaged for 1991 and 1992. **, * Mean in DF is significantly less than the control at the 0.05 and 0.01 probability levels, respectively.

(Table 1) and was correlated with pod number ($r = 0.61$, $P = 0.0001$) and yield ($r = 0.61$, $P = 0.0001$). Yield losses in the more severe defoliation treatments (DF-R5 and DF-R6.5) were related not only to a 33% decrease in dry matter at R7 (data not shown) but also to a 5% decrease in the fraction of dry matter partitioned into yield.

Reduced pod production played a prominent role in decreasing yield. Significant seed size reductions occurred for the DF-R5 and DF-R6.5 treatments in the current study. However, seed size never accounted for more than 25% of yield loss (Table 1) and was not significantly correlated with yield.

Both main stem and branch pod number and yields were significantly less in defoliation treatments vs. control (Table 3 and 4). Reduced main stem pod number was entirely due to fewer pods per reproductive node ($r = 0.87$, $P = 0.0001$). However, decreased branch

pod production resulted mainly from fewer reproductive nodes ($r = 0.78$, $P = 0.0001$) and, to a lesser degree, fewer pods per reproductive node ($r = 0.64$, $P = 0.0001$). Decreased branch reproductive node number was directly related to fewer nodes ($r = 0.88$, $P = 0.0001$), branches ($r = 0.75$, $P = 0.0001$), and branch dry matter ($r = 0.48$, $P = 0.01$). Results differ from previous research (Board and Harville, 1993) where reproductive node number was unaffected by defoliation treatment. In that study, LI differences between control and defoliation treatments were not apparent until R3 in the first year. Although greater LI differences occurred in the second year, they were not great enough to affect reproductive node number. In contrast, LI differences in the current study between defoliation treatments vs. control (Fig. 1) were consistently large enough during R1 to R5 to significantly reduce reproductive nodes.

Table 1. Seed yield, yield components, and harvest index for control and defoliation treatments (DF), averaged for 1991 and 1992, for soybean planted at Baton Rouge, LA.

Defoliation treatment	Seed yield			Pod number	Seed per pod	Seed number	Seed size	Harvest index	Reduction in yield due to	
	1991	1992	Mean						Reduced seed no.	Reduced seed size
	kg ha ⁻¹			no. m ⁻²	no.	no. m ⁻²	g per 100 seed	%	%	
Control	3592	2644	3118	1593	2.14	3396	11.66	59		
DF-R3	2831**	2559	2695**	1351**	2.12	2870**	11.48	60	100	0
DF-R4	2734**	2306*	2520**	1195**	2.10	2515**	11.25	57**	100	0
DF-R5	2307**	2235*	2271**	1084**	2.02	2266**	11.04*	56**	86	14
DF-R6.5	2095**	2082**	2091**	1162**	2.02	2360**	10.55**	56**	76	24

*, ** Means in DF are significantly less than the control at the 0.05 and 0.01 probability levels, respectively.

Table 2. Anovar for seed yield, yield components, and harvest index for soybean planted at Baton Rouge, LA, 1991–1992.

Anovar source	Summary df†	Seed yield	Pod number	Seed per pod	Seed number	Seed size	Harvest index
Year (Y)	1	**	**	NS	**	**	NS
Defoliation treatment (DF)	4	**	**	NS	**	**	**
Y × DF	4	**	NS	NS	NS	NS	NS
CV, %		8.4	9.3	4.6	10.7	4.0	2.5

** Mean square is significant at the 0.01 probability level; NS = not significant.

† df = degrees of freedom.

Pod Appearance and Abortion

Pods per reproductive node for each DF–DAE mean were averaged across years and presented in Fig. 2. Regardless of treatment, maximal pods per reproductive node occurred between R4 to R5. Thus, pod initiation occurred during a 1-wk period between R3 and R5. Maximal pods per reproductive node were significantly reduced in all defoliation treatments vs. control (Table 5). However, all defoliation treatments had similar maximal pods per reproductive node. Flower production probably did not influence these changes, because studies using shade cloth that reduced LI 30% (comparative to reduced LI created by defoliation in the current study) showed no effect on flower production in soybean (Jiang and Egli, 1993).

Immediately after R5, production of new pods stopped and pod number began to decline (Fig. 2). The largest drop occurred between 59 to 67 DAE (5 to 13 d after R5). In all treatments, little change in pods per reproductive node occurred beyond this point. Thus, determination of final pod number occurred at 10 to 12 d after R5. Pod formation demonstrated a pod initiation period, R3 to R5, during which all potential viable pods were initiated, and a pod abortion period, R5 to 10 to 12 d after R5, in which a certain percentage of initiated pods (all <2 cm in length) failed to develop into viable pods. Significant differences in final pods per reproductive node occurred between defoliation treatments vs. control, as well as among defoliation treatments themselves (Table 5).

As defoliation treatments extended up to and past R5 (DF-R4, DF-R5, and DF-R6.5), significantly greater abortion occurred compared with the control (Table 5). Control plants aborted about 28% of maximal pods per reproductive node, even with 97% LI (Fig. 1). Pod abortion was similar in DF-R3 vs. control, probably due to the high LI (85–90%) during the pod abortion period for DF-R3 (Fig. 1). Reduced pods per reproductive node

for DF-R3 vs. control were entirely due to reduced pod initiation. Significantly greater pod abortion compared with control occurred for all other defoliation treatments (Table 5), because of lower LI (75–80%) prevailing during pod abortion (Fig. 1). Reduced pods per reproductive node for these treatments vs. control were half due to reduced pod initiation and half due to increased pod abortion. Thus, sensitivity of pods per reproductive node to LI appears regulated through effects on pod initiation and pod abortion. Near-optimal LI needs to be maintained into the first 10 to 12 d of the seed filling period to maximize pods per reproductive node.

Implications for Yield Formation

Previous yield formation studies have concentrated on source strength effects on reproductive events such as seed growth rate (Egli et al., 1985), length of the seed filling period (Egli et al., 1985), and flower–pod abortion (Jiang and Egli, 1993). Such studies are useful in explaining environmental effects on yield through source strength effects on these parameters. Thus, farmers can be advised when optimum source strength is necessary for highest yield. However, seed growth rate has shown little relationship to environmental effects on yield (Egli, 1975). Similarly, length of effective filling period has not shown a strong relationship to yield across field environments (Salado-Navarro et al., 1986). Flower production and flower–pod abscission have been shown to respond only to drastic reductions (63%) in source strength which do not frequently occur under natural conditions (Jiang and Egli, 1993).

Results from the current study indicate that source strength affects pod number and yield through mechanisms different from those explored in previous studies. Results of the current study demonstrate that source strength affects branch number and node number, as well as pods per reproductive node on both main stems

Table 3. Branch and main stem yield and yield components for control and defoliation treatments (DF), averaged for 1991 and 1992, for soybean planted at Baton Rouge, LA.

Defoliation treatment	Main stem						Branch stem					
	Yield	Pods per reproductive node	Pod number	Yield	Dry matter	Total length	Number	Node number	Percentage reproductive nodes	Reproductive node number	Pods per reproductive node	Pod number
	g m ⁻²	no.	no. m ⁻²	g m ⁻²	g m ⁻²	cm m ⁻²	no. m ⁻²	no. m ⁻²	%	no. m ⁻²	no.	no. m ⁻²
Control	182	2.97	721	212	29.0	2572	275	581	86	499	1.75	872
DF-R3	158**	2.65*	624**	170**	20.5**	1929**	251	527	87	452	1.63*	727**
DF-R4	153**	2.50**	616**	129**	15.9**	1668**	220**	479**	83	392**	1.49**	578**
DF-R5	135**	2.21**	558**	115**	16.6**	1838**	227**	479**	77**	362**	1.47**	525**
DF-R6.5	127**	2.52**	568**	119**	16.7**	1776**	226**	478**	80**	379**	1.58**	594**

*, ** Means in DF are significantly less than the control at the 0.05 and 0.01 probability levels, respectively.

Table 4. Anovar for branch and main stem yield components for soybean planted at Baton Rouge, LA, 1991-1992.

Anovar source	Summary dft	Main stem			Branch stem								
		Yield	Pods per reproductive node	Pod number	Yield	Dry matter	Total length	Number	Node number	Percentage reproductive nodes	Reproductive node number	Pods per reproductive node	Pod number
Year (Y)	1	**	**	**	NS	**	**	**	*	**	NS	**	*
Defoliation treatment (DF)	4	**	**	**	**	**	**	**	*	**	**	**	**
Y × DF	4	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	NS	**
CV, %		9.2	10.2	11.1	16.4	25.4	23.5	11.3	13.9	5.3	12.6	6.8	12.7

*, ** Mean square is significant at the 0.05 and 0.01 probability levels, respectively; NS = not significant.

† df = degrees of freedom.

and branches. The current study also shows that pods per reproductive node were controlled both by pod initiation and abortion. Thus, source restriction is evident from R1 to determination of final pod number (10 to 12 d after R5). Because the study was conducted at a late planting, this yield formation mechanism may not be operative at an optimal planting date. However, research involving 30% shade (R1 to R7) applied at an optimal planting date indicates a similar yield formation response (unpublished data). Results from the current study should not be interpreted to mean that seed per pod and seed size are completely independent of source strength. Future studies are needed to address this issue. The purpose of the current study was to identify when pod number was source restricted.

Source restriction for pod number during R1 through

10 to 12 d after R5 helps explain why many studies have identified R1 to R5.5 as the most sensitive to environmental stress. Elmore et al. (1988) cited several surface-irrigation studies demonstrating the importance of adequate soil moisture during R3 to R4 for best yield. Defoliation-induced yield losses have been greatest at R4, R5, or R5.5 compared with other growth stages (Fehr et al., 1981; Goli and Weaver, 1986). Schou et al. (1978) reported greatest yield response to supplemental light between late flowering and early pod formation. Simulated lodging studies (Woods and Swearingin, 1977) indicated that R5 was the most sensitive stage. Yield losses at nonoptimal planting dates (Board, 1985) and nonoptimal row spacings (Board et al., 1990, 1992) occur through effects on yield components formed during R1 to R5.5. In conclusion, optimum source strength

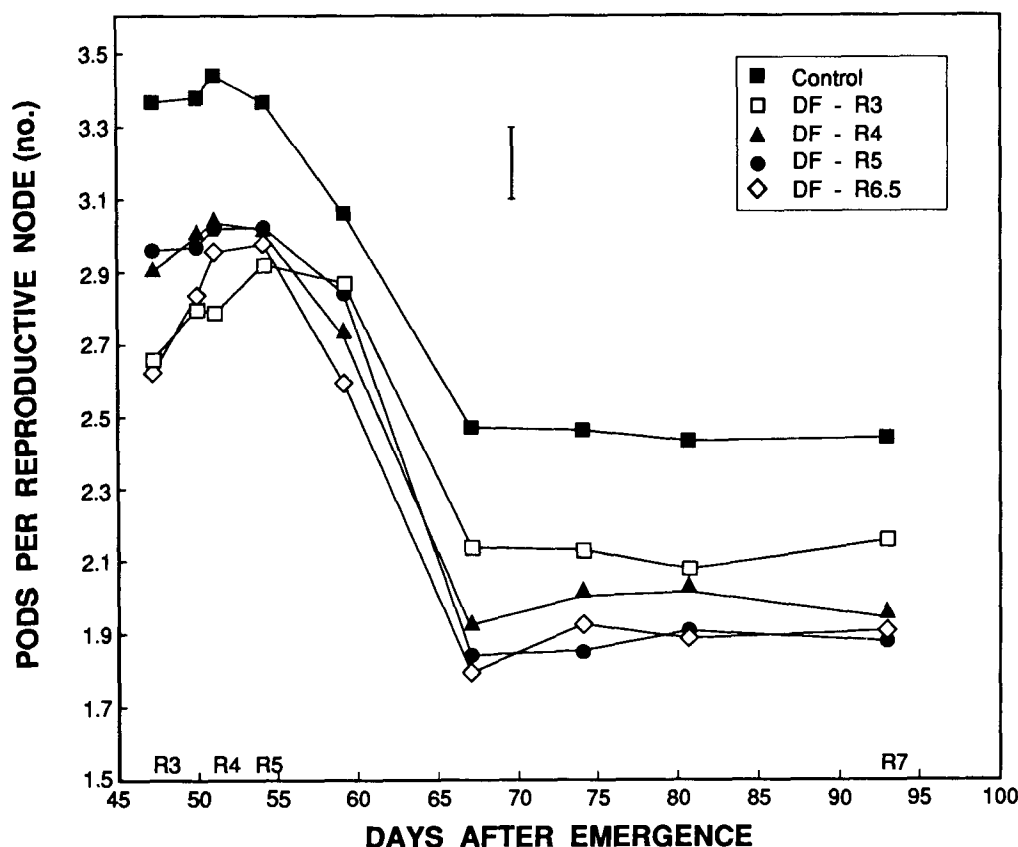


Fig. 2. Pods per reproductive node between pod initiation (R3) and physiological maturity (R7) as affected by defoliation treatments (DF), averaged for 1991 and 1992 [vertical line represents LSD ($P = 0.05$) for comparing pods per reproductive node between sampling dates within treatments or between treatments within a sampling date].

Table 5. Maximal pod number, final pod number, and percentage aborted pods of control and defoliation treatments (DF) for soybean averaged for 1991 and 1992, Baton Rouge, LA.

Defoliation treatment	Pods per reproductive node		Reduced pods due to		
	Maximal	Final	Pod abortion	Decreased pod initiation	Increased pod abortion
	no.	no.	%	%	%
Control	3.41	2.45	28		
DF-R3	2.86**	2.13**	26	100	0
DF-R4	3.03**	1.99**	34†	59	41
DF-R5	3.02**	1.87**	38**	48	52
DF-R6.5	3.00**	1.87**	38**	50	50

** Means in DF are significantly different from control at the 0.01 probability level.

† Mean in DF is significantly higher than control at the 0.08 probability level.

should be maintained during R1 to shortly after R5 to achieve maximum pod number and yield.

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